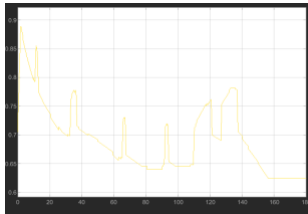


Software Engineering Data Processing UAV AI

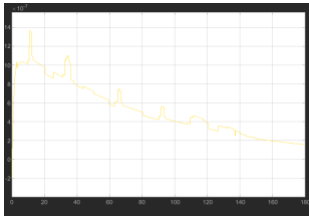
Case- No disturbances on centerline, holding a minimum mission Altitude.

Scopes were added on command inputs

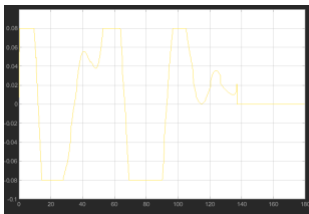
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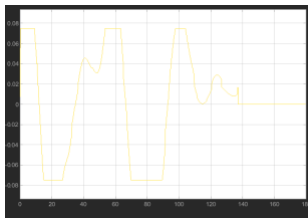
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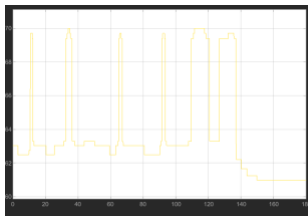
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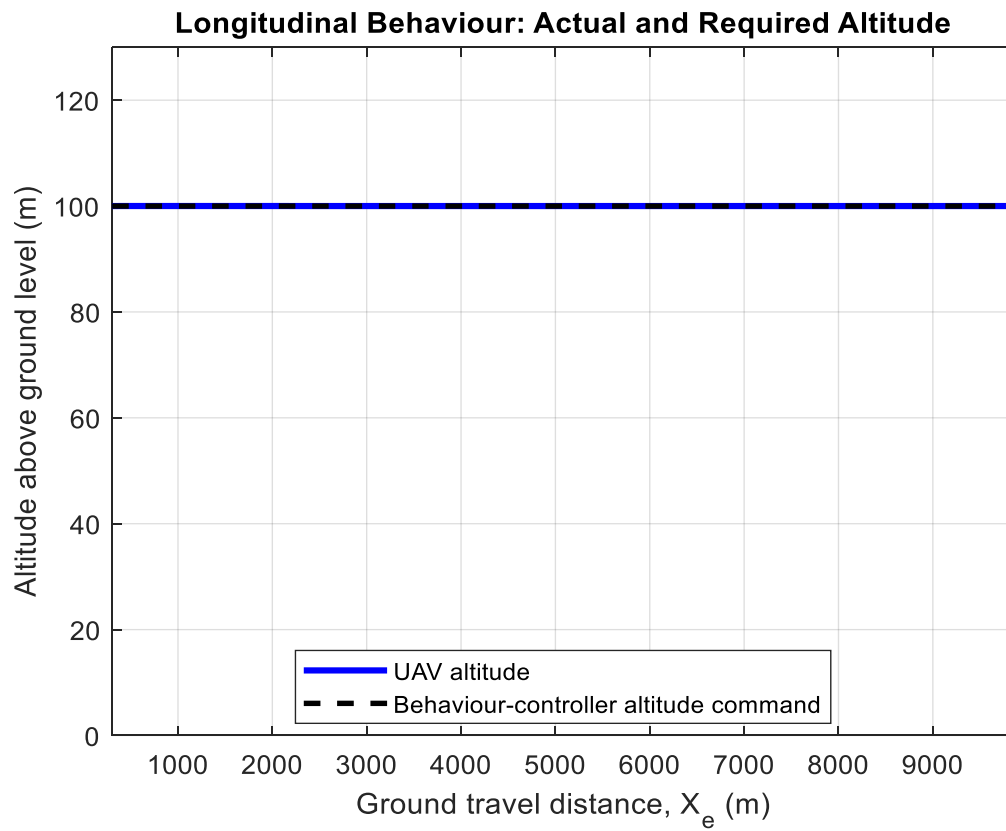
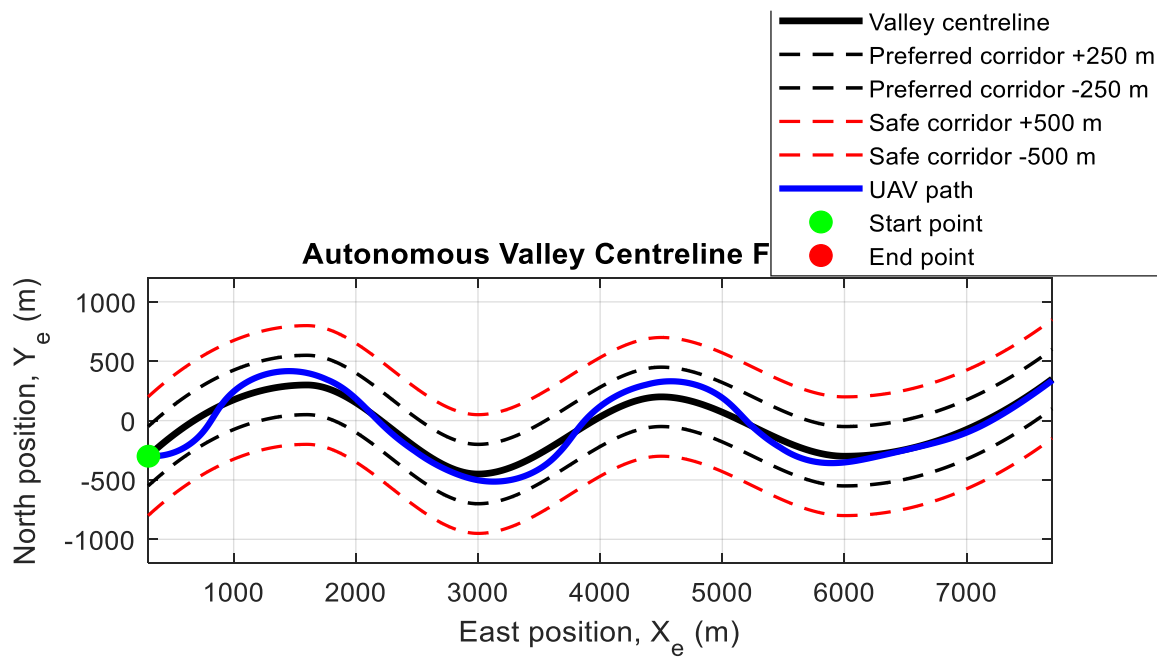


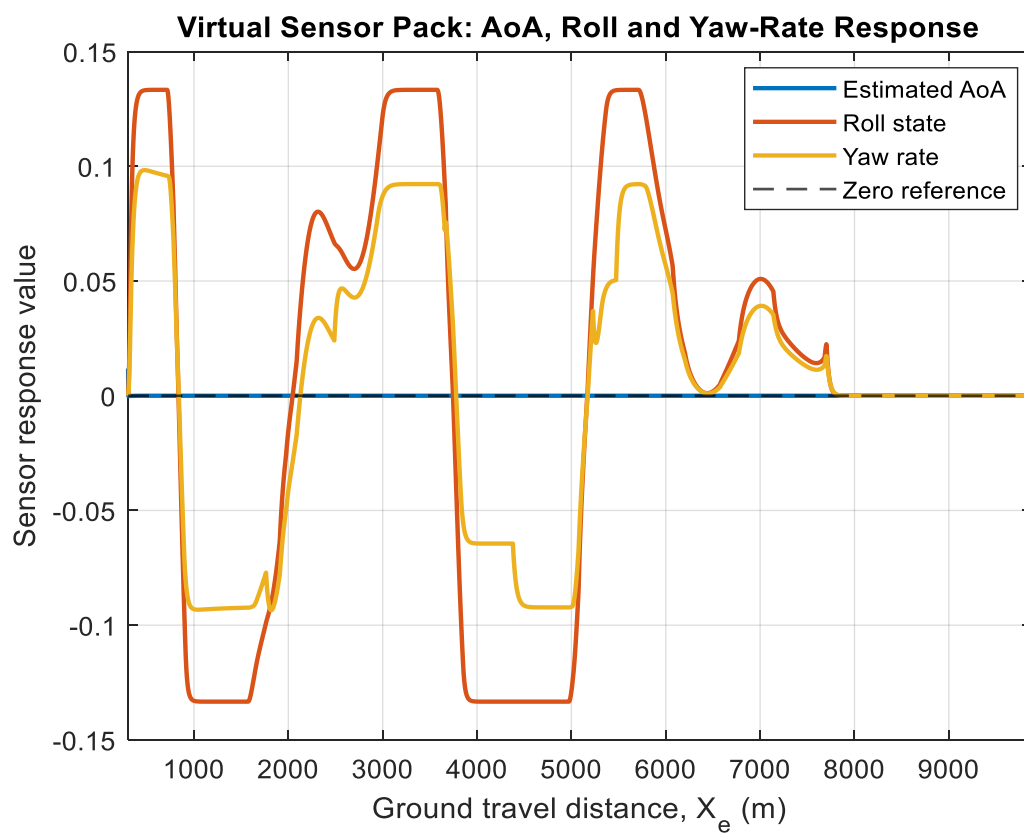
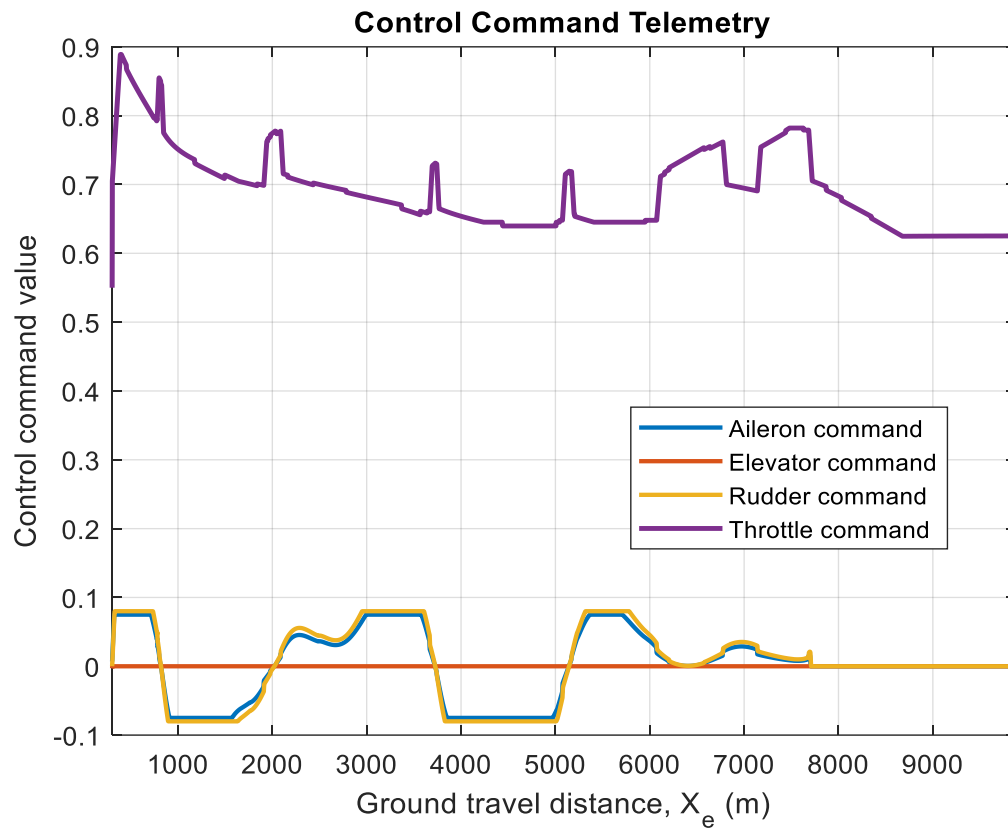
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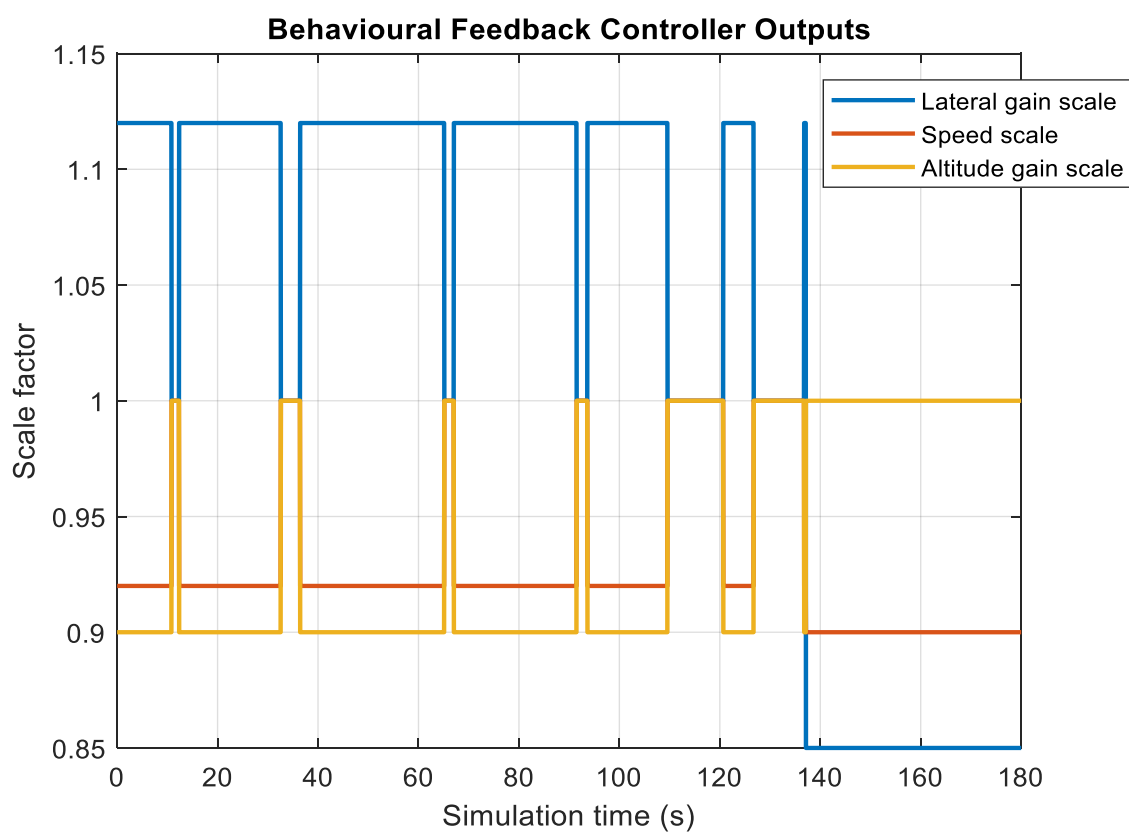
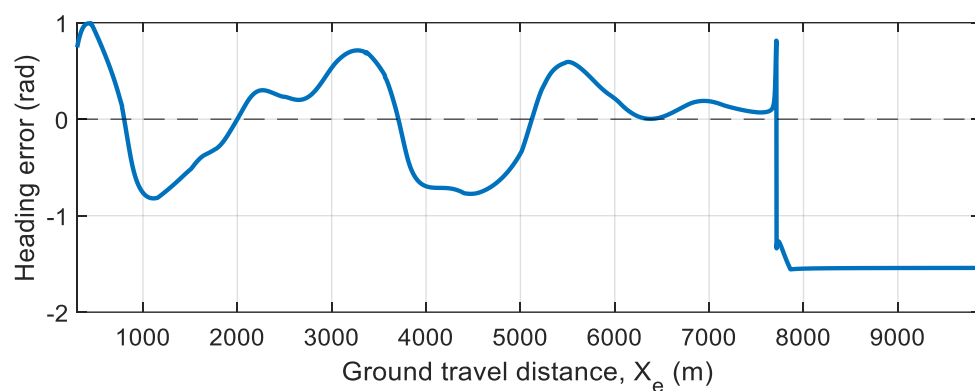
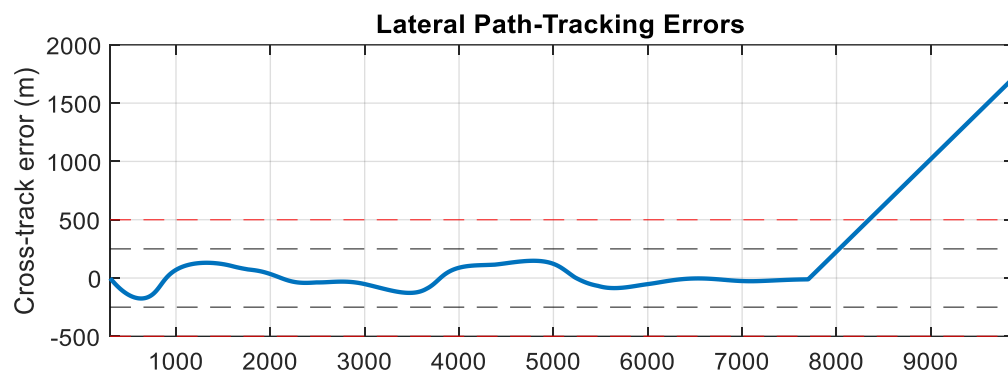


Speed









Baseline Results: Validation of Nominal Mission Requirements

The baseline simulation was conducted with the centreline disturbance input disabled in order to evaluate the nominal performance of the proposed autonomous UAV guidance and control architecture. This case represents the primary mission condition in which the aircraft is required to navigate through a mapped valley, remain within defined lateral operating limits, maintain the scheduled mission altitude, and complete the route without direct pilot intervention.

The mission concept requires the UAV to use the mapped valley centreline as a guidance reference rather than as an exact geometric path. A preferred lateral corridor of ± 250 m and a maximum safe corridor of ± 500 m were therefore defined around the reference trajectory. This requirement reflects the practical need to permit bounded tracking error while preventing the aircraft from deviating excessively toward surrounding terrain.

The simulated trajectory followed the general curvature of the valley from the initial position to the final waypoint. The UAV remained within the ± 500 m safe corridor throughout the nominal mission and operated predominantly within the ± 250 m preferred corridor. The observed response therefore satisfies the principal lateral-navigation requirement of maintaining controlled flight inside the mapped valley limits. The small differences between the UAV trajectory and the reference centreline are consistent with a dynamic aircraft response and do not represent guidance failure, because the centreline was defined as a reference path rather than a mandatory zero-error trajectory.

The aileron and rudder commands changed direction in response to the alternating curvature of the valley route. Their magnitudes remained bounded at approximately ± 0.08 , indicating that the lateral controller generated finite and controlled commands rather than unstable or continuously increasing corrections. The command reversals corresponded to the major left- and right-hand path segments, while the reduction in command magnitude after the final major turn indicated convergence toward the end-of-mission condition. This behaviour validates the design concept of using heading error and cross-track error to generate coordinated lateral corrections.

The longitudinal mission requirement was to maintain a scheduled altitude of 100 m above ground level during nominal valley-following flight. The actual UAV altitude remained closely aligned with the commanded altitude over the complete route. Only a small gradual deviation was observed toward the end of the simulation, and the aircraft remained near the required 100 m level. The elevator command remained very small, showing that the longitudinal controller required only limited corrective input once the altitude-hold condition had been established. This result validates the altitude-hold function under the nominal operating case.

The speed-control requirement was to maintain a stable forward velocity while allowing limited adjustment during changes in lateral workload. The simulated speed remained approximately within the range of 61–70 m/s. Corresponding throttle variations remained bounded between approximately 0.62 and 0.89. Temporary increases in throttle occurred during periods of increased manoeuvring demand, while the command reduced as the aircraft progressed toward route completion. This confirms that the propulsion command was actively adjusted to support the flight path without producing unbounded acceleration or loss of speed control.

The disturbance-detection signal remained at zero throughout the run, confirming that no artificial centreline obstruction was active. Under this condition, the behavioural controller did not initiate the climb-priority or direct-escape response. Instead, the behaviour mode alternated primarily between normal tracking and lateral recovery. These changes occurred when the aircraft required temporary correction of heading or cross-track error and are consistent with the intended supervisory logic. The final transition to the mission-complete state confirmed that the system recognised completion of the mapped route.

The virtual sensor outputs provide an additional assessment of aircraft response. The estimated angle of attack remained close to zero, while roll and yaw-rate responses followed the required lateral manoeuvres and returned toward zero after the final path segment. The responses remained bounded throughout the mission, indicating that the control commands did not produce divergent or excessively oscillatory motion within the simplified three-degree-of-freedom model.

Overall, the baseline simulation satisfies the principal nominal mission requirements:

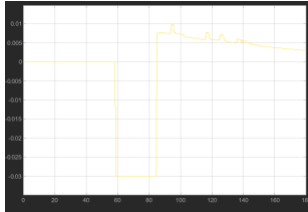
1. autonomous traversal of the mapped valley route;
2. maintenance of the UAV within the ± 500 m safe corridor;
3. predominant operation within the ± 250 m preferred corridor;
4. regulation of altitude near the scheduled 100 m mission level;
5. bounded speed and control-command response;
6. correct operation of the normal-tracking and lateral-recovery behavioural states;
and
7. successful recognition of mission completion.

These results validate the proposed concept of design under nominal conditions. In particular, they demonstrate that the environment map, lateral guidance controller, longitudinal altitude controller, aircraft model, virtual sensor pack, and behavioural supervisor can operate together without the supervisory additions degrading the underlying path-following performance. The baseline case therefore establishes a valid

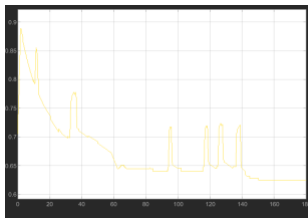
reference against which the disturbance-response and climb-priority cases can be assessed.

Case-Disturbances on centerline

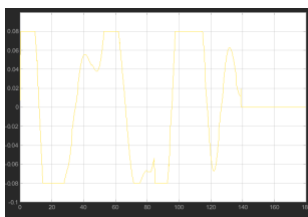
Elevator



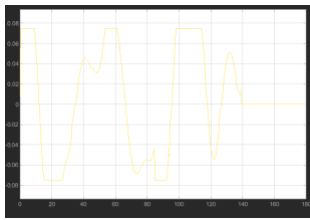
Throttle



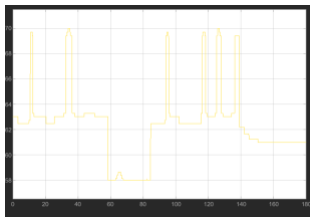
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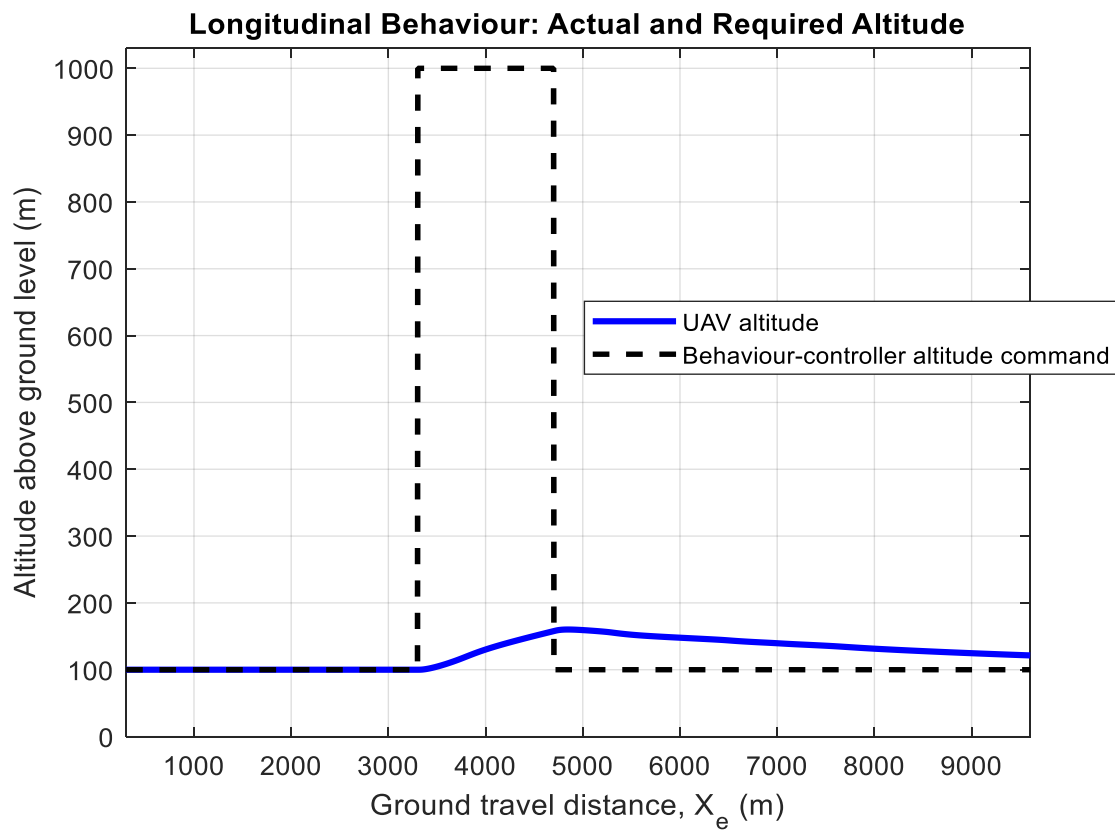
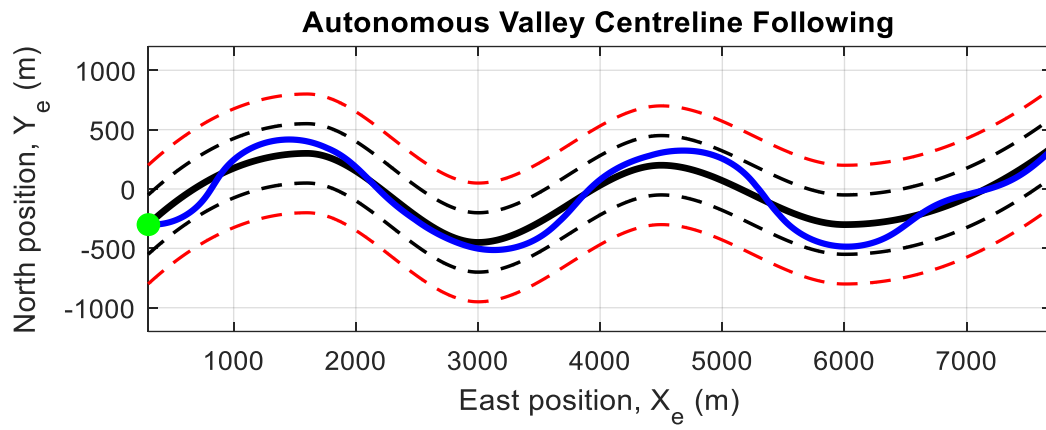


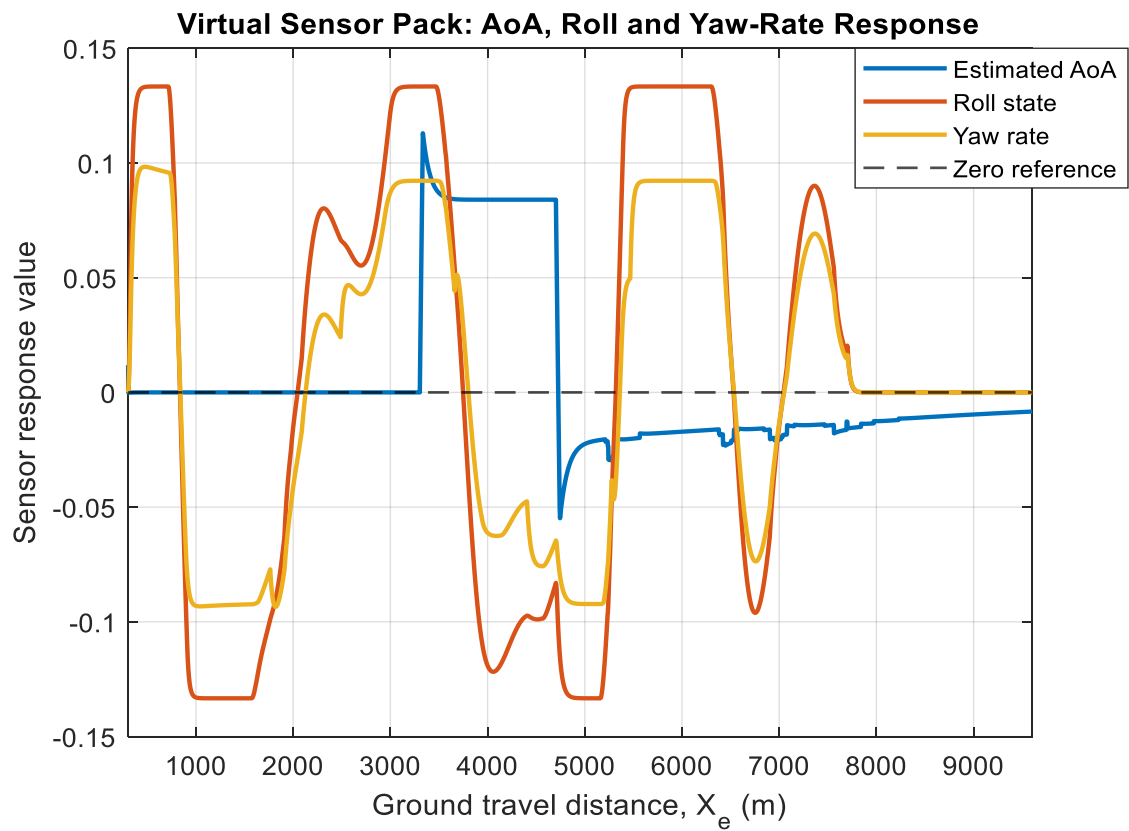
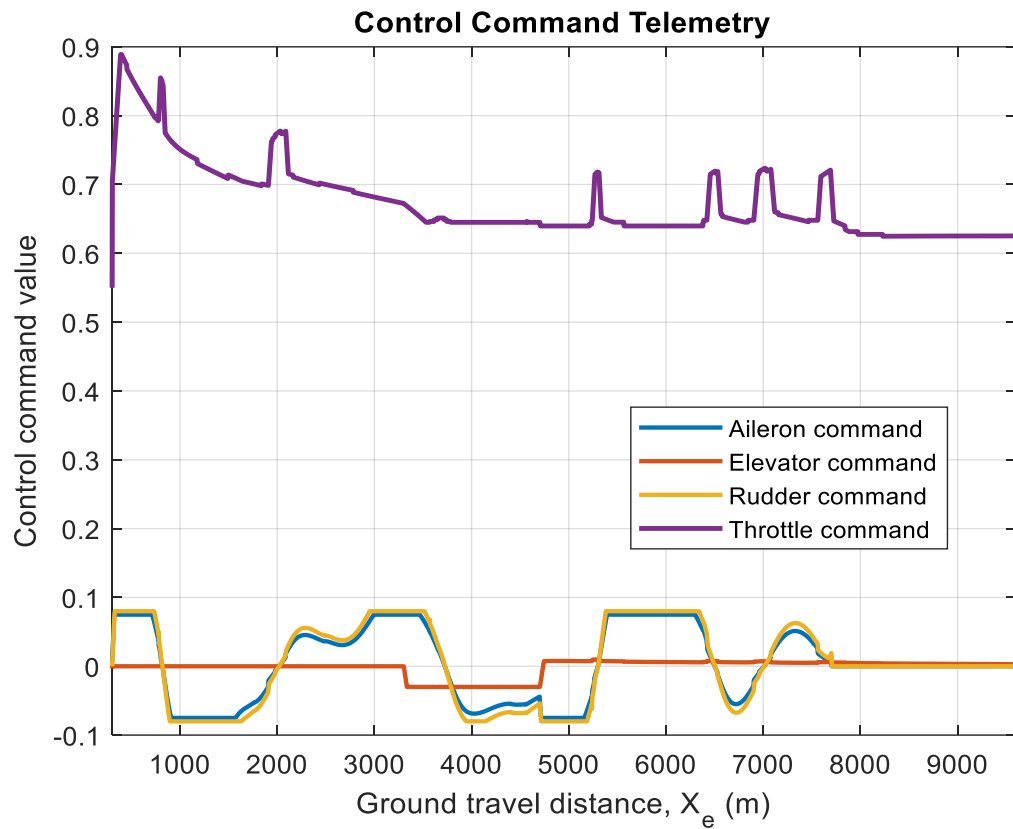
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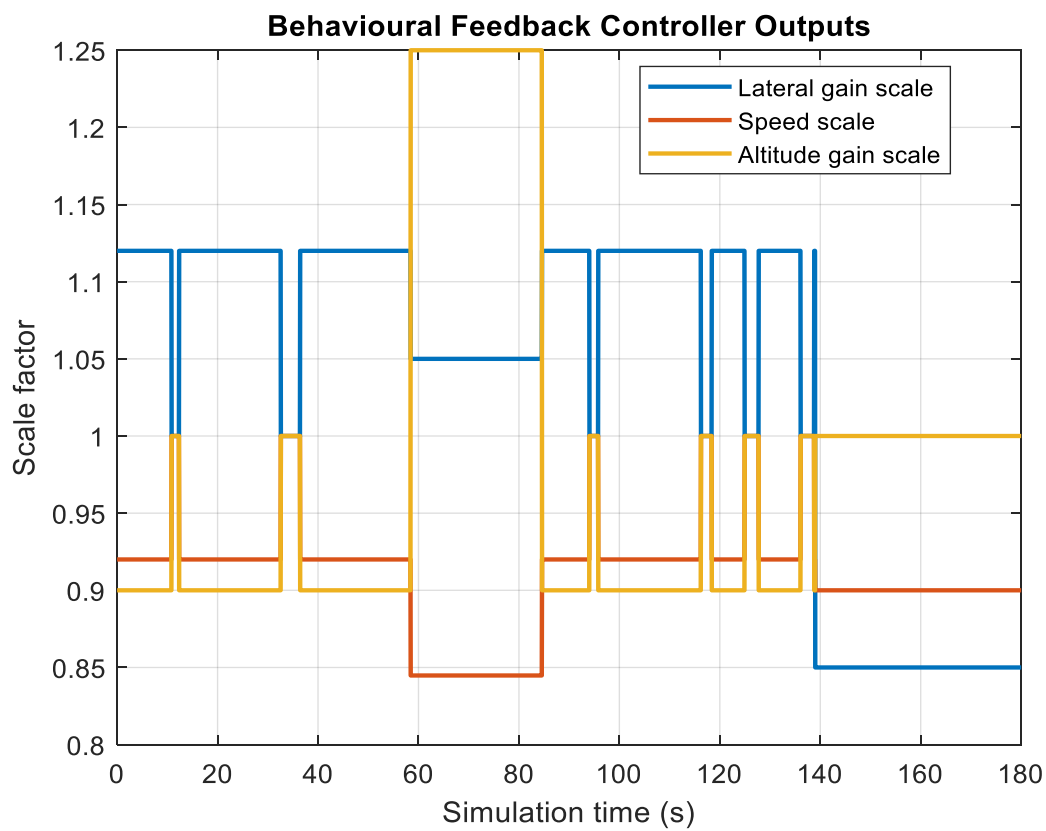
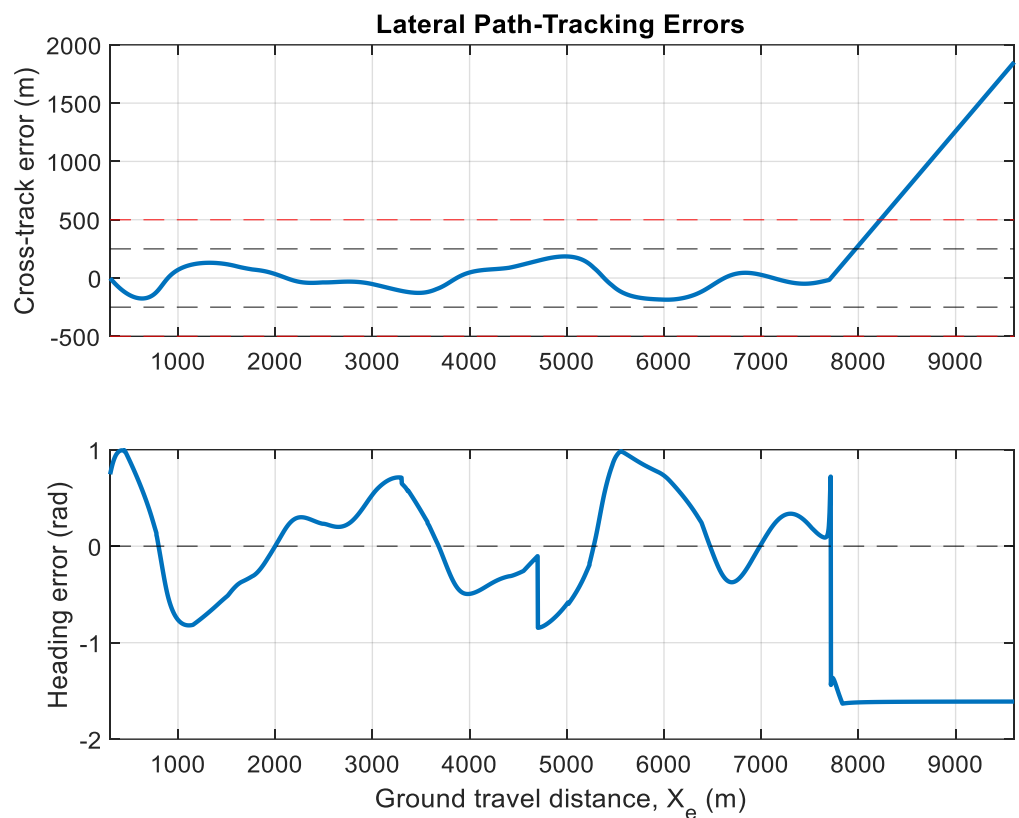


Speed









Disturbance Case: Behavioural Response and Concept Validation

The second simulation case evaluated the UAV under a predefined centreline disturbance condition with the disturbance input set to level 2. The objective of this test was to demonstrate whether the proposed behavioural-control architecture could detect a hazardous map region, change the UAV's operating priorities, initiate a climb response, maintain lateral containment within the valley corridor, and return toward nominal mission behaviour after the disturbance cleared.

The disturbance was first detected at 58.45 s, corresponding to an east position of approximately 3302.97 m. It remained active until 84.55 s, producing a disturbance interval of 26.10 s. During this period, the disturbance-detection signal changed from zero to one and the behavioural controller transitioned into the disturbance/climb-priority mode. This response demonstrates that the environment-map logic and supervisory behaviour layer operated together to recognise the imposed hazard and modify the mission response.

Under nominal conditions, the UAV was commanded to maintain an altitude of 100 m above ground level. When the disturbance was detected, the behavioural controller increased the required altitude command to 1000 m. This was a deliberately large and hypothetical emergency command used to demonstrate that the supervisory layer could override the nominal mission-altitude objective when a hazard was detected.

The behavioural controller simultaneously increased the altitude gain scale to approximately 1.25 and reduced the speed scale to approximately 0.85. Lateral control authority was retained so that the UAV could continue tracking the valley route while responding vertically. This represents the intended behavioural prioritisation strategy: increase climb demand, reduce forward-speed demand, and preserve sufficient lateral stability during the disturbance response.

The actual UAV altitude increased from 100 m to a maximum of 157.73 m during the active disturbance interval. The maximum altitude reached over the complete simulation was 160.16 m. Although the supervisory altitude command was set to 1000 m, the aircraft response remained governed by its simulated kinematics and dynamics, including available elevator authority, throttle response, forward speed, inertia, flight-path angle, and achievable climb rate. The UAV therefore responded progressively rather than instantaneously, producing a stable climb within the available 26.10 s disturbance interval.

This difference between the commanded and achieved altitude demonstrates the separation between high-level behavioural control and low-level aircraft response. The behavioural controller generated the emergency mission objective, while the longitudinal controller and UAV plant determined the physically achievable motion. The

aircraft did not directly jump to the commanded altitude; instead, it produced a bounded climb response consistent with the modelled actuator and dynamic limitations.

Once the disturbance cleared, the required altitude returned to the nominal 100 m mission level. The UAV then descended gradually and settled near 120 m by the end of the simulation. The remaining difference between the final altitude and the nominal command is consistent with the transient response of the aircraft after the climb phase. At the end of the disturbance interval, the UAV retained positive vertical motion and residual flight-path response, requiring time to return fully to the original altitude condition.

This behaviour demonstrates the intended role of the AI-based supervisory controller within the conceptual prototype. Rather than continuing to follow the nominal altitude and speed commands, the controller interpreted the disturbance state and reprioritised the UAV response by increasing altitude demand, reducing speed demand, and maintaining lateral stability. The resulting aircraft motion remained governed by the kinematic and dynamic characteristics of the simulated UAV model.

Lateral guidance remained effective during the disturbance response. The maximum cross-track error recorded during the active disturbance interval was 147.43 m. This remained within the preferred ± 250 m corridor and well within the maximum safe corridor of ± 500 m. The UAV therefore maintained controlled valley tracking while performing the climb-priority manoeuvre, demonstrating that the longitudinal response did not cause a significant loss of lateral containment.

The control-command outputs remained bounded throughout the simulation. The maximum absolute aileron, elevator and rudder commands were 0.075, 0.030 and 0.080 respectively. Throttle remained between 0.550 and 0.889. These values show that the disturbance response was achieved through finite and controlled actuator commands. The elevator response changed clearly during disturbance entry and subsequently reversed as the UAV returned toward the nominal altitude.

Aircraft speed varied between 45.03 and 65.00 m/s, with a mean value of 63.90 m/s. The temporary reduction in speed during the disturbance interval was consistent with the reduced speed scale and the increased longitudinal demand associated with the climb response. The UAV later recovered toward the nominal operating-speed range after the disturbance cleared.

The virtual sensor outputs also remained bounded. The maximum estimated angle of attack was 0.11296 rad, the maximum roll magnitude was 0.13333 rad, and the

maximum yaw-rate magnitude was 0.09832 rad/s. The sensor-status code remained at 1 throughout the manoeuvre, indicating that the defined virtual sensor limits were not exceeded.

The behaviour-mode distribution further confirms the operation of the supervisory logic. The UAV spent 8.19% of the simulation in normal tracking, 14.52% in disturbance/climb-priority mode, 54.48% in lateral recovery, and 22.80% in mission-complete mode. The disturbance/climb-priority mode aligned with the detected hazard interval, while the later transition to mission-complete mode confirmed successful recognition of the end of the mapped route.

The full-run summary reported a maximum absolute cross-track error of 1850.89 m. This large value occurred after the UAV had passed the final mapped waypoint at approximately 7700 m while the simulation continued to propagate the aircraft to an east position of 9602.65 m. The mission-valid disturbance performance is therefore represented more accurately by the event-specific cross-track error of 147.43 m recorded while the UAV remained within the defined map domain.

Overall, the disturbance case met the principal concept-design objectives. The system:

- detected the predefined centreline disturbance;
- transitioned into a disturbance-response operating mode;
- increased the required altitude command;
- increased altitude-control priority;
- reduced speed demand;
- initiated a stable climb response;
- maintained lateral containment within the preferred corridor during the disturbance;
- retained bounded actuator and sensor behaviour;
- returned toward nominal altitude after the disturbance cleared; and
- completed the mapped mission.

The results validate the proposed behavioural-control concept at the integrated simulation level. The environment map, supervisory behaviour controller, lateral guidance system, longitudinal controller, custom UAV model, and virtual sensor pack operated together to detect a hazard, reprioritise the mission response, and maintain controlled autonomous flight.

Comparative Evaluation of Nominal and Disturbance Cases

The two simulation cases were used to evaluate the proposed autonomous UAV guidance architecture under both nominal and disturbed operating conditions. The baseline case assessed whether the UAV could follow the mapped valley route, remain within the defined lateral corridor, maintain the nominal mission altitude, and complete the route without a disturbance response. The second case assessed whether the same architecture could detect a predefined centreline disturbance, change behavioural priorities, initiate a climb response, preserve lateral containment, and recover toward nominal operation.

Nominal Case

With the centreline disturbance input disabled, the UAV followed the curved valley route while maintaining the scheduled altitude of approximately 100 m above ground level. The trajectory remained within the ± 500 m safe corridor throughout the valid mission and operated predominantly within the preferred ± 250 m corridor.

The disturbance-detection signal remained inactive, and the behavioural controller operated mainly between normal tracking and lateral-recovery modes. The lateral-recovery state was activated when additional heading or cross-track correction was required during curved sections of the route. The system later transitioned to mission-complete mode after reaching the end of the mapped path.

The actual altitude remained close to the 100 m command, while the elevator command remained small. This indicated that the longitudinal controller required only limited correction once the nominal altitude-hold condition had been established. Speed and actuator commands also remained bounded, demonstrating stable operation of the integrated lateral and longitudinal control system.

This result confirmed that the addition of the behavioural controller, virtual sensor pack, disturbance logic, and supervisory gain scheduling did not degrade the underlying nominal guidance response.

Disturbance Case

With the centreline disturbance input set to level 2, the disturbance was detected at 58.45 s and remained active for 26.10 s. The behavioural controller responded by transitioning into disturbance/climb-priority mode, increasing the required altitude from 100 m to the hypothetical emergency command of 1000 m, increasing the altitude-control gain, and reducing the speed scale.

The UAV climbed from approximately 100 m to a maximum of 157.73 m during the active disturbance period and reached a maximum altitude of 160.16 m over the complete simulation. The difference between the 1000 m supervisory command and the achieved altitude reflects the dynamic response of the aircraft model. The high-level controller changed the mission objective, while the elevator, throttle, airspeed, flight-path angle, inertia, and available climb duration determined the achievable physical response.

After the disturbance cleared, the altitude command returned to 100 m. The UAV subsequently descended and settled near 120 m by the end of the simulation. This gradual return reflected the transient response following the climb phase rather than an instantaneous transition between altitude commands.

The maximum cross-track error during the actual disturbance interval was 147.43 m. This remained within the preferred ± 250 m corridor and well within the ± 500 m safe corridor. The UAV therefore maintained lateral route containment while prioritising the climb response.

Actuator commands and virtual sensor outputs remained bounded during the disturbance event. The maximum absolute aileron, elevator, and rudder commands were 0.075, 0.030, and 0.080 respectively, while throttle remained between 0.550 and 0.889. Angle of attack, roll, and yaw-rate responses also remained bounded, and the sensor-status output remained in the normal state.

Comparison of the Two Cases

The nominal and disturbance simulations produced clearly different behaviours while using the same core UAV model and control architecture.

In the nominal case, the environment remained clear, the disturbance-detection signal stayed at zero, and the behavioural controller maintained the normal mission objective. The UAV followed the curved valley route, remained within the defined corridor limits, and held the scheduled altitude of approximately 100 m.

In the disturbance case, the environment-map input activated a predefined hazard region. The disturbance-detection signal changed from zero to one, and the behavioural controller switched into climb-priority mode. The required altitude increased from 100 m to 1000 m, the altitude-control gain increased, and the speed scale reduced. The UAV responded by climbing to approximately 160 m while continuing to track the valley route within the preferred corridor during the active disturbance interval.

The difference between the two cases therefore demonstrates the purpose of the behavioural layer. Under nominal conditions, the system prioritised path tracking and altitude hold. Under disturbed conditions, it changed the mission priorities by increasing altitude demand and reducing speed demand while retaining lateral guidance authority.

The disturbance case also demonstrated that the aircraft response remained bounded despite the large supervisory altitude command. The UAV did not attempt an instantaneous jump to 1000 m. Instead, the low-level controller and aircraft dynamics produced a progressive climb consistent with the available actuator authority and flight conditions. After the disturbance cleared, the UAV gradually returned toward the nominal altitude.

This behaviour indicates that the supervisory controller did not replace the lower-level flight dynamics. It modified the mission objective, while the aircraft model determined the achievable motion. This separation between high-level behavioural decision-making and low-level dynamic response is an important feature of the proposed architecture.

The lateral response also remained effective in both cases. In the nominal simulation, the UAV remained within the safe corridor throughout the valid route. In the disturbance simulation, the maximum cross-track error during the active event was 147.43 m, which remained inside the preferred ± 250 m corridor. The climb-priority response therefore did not cause a loss of lateral containment.

The actuator and sensor results were also consistent across both cases. Aileron, elevator, rudder, and throttle commands remained finite and bounded. The angle-of-attack, roll, and yaw-rate responses remained controlled, and the virtual sensor-status output remained in the normal state. This demonstrates that the addition of behavioural switching did not destabilise the aircraft model.

Overall Results Summary

The combined results demonstrate the feasibility of the proposed map-informed autonomous UAV guidance concept.

The nominal case showed that the UAV could autonomously follow a curved valley route, remain within the defined lateral operating limits, hold the scheduled mission altitude, regulate forward speed, and complete the mapped mission without direct pilot steering.

The disturbance case showed that the same architecture could detect a predefined environmental hazard, change behavioural mode, override the nominal altitude objective, increase altitude-control priority, reduce speed demand, initiate a stable climb response, and preserve lateral tracking during the event.

Together, the two cases validate the central concept of design. The environment map, behavioural supervisor, lateral controller, longitudinal controller, custom UAV model, gust model, and virtual sensor pack operated as an integrated system.

The baseline case confirmed that the additional behavioural and monitoring functions did not degrade the nominal guidance response. The disturbance case confirmed that the system could modify its operating priorities when the environment changed.

The principal achievement of the design is therefore not only that the UAV can follow a mapped route, but that it can alter its behaviour according to the interpreted mission condition. This supports the overall mission objective of developing a conceptual autonomous UAV system capable of map-based navigation, altitude control, disturbance response, behavioural mode selection, and operator-visible telemetry without continuous pilot input.